

YIXUAN ASHER ZHANG

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EDUCATION

University of Illinois Urbana-Champaign

Master of Science in Computer Science

Champaign, IL

May 2028 (expected)

Coursework: Machine Learning Systems, Distributed Systems, Advanced Operating Systems, Advanced Topics in NLP, Computer Networks, Advanced Computer Security.

University of Nottingham

B.Sc. (Hons) Computer Science, First Class · GPA 3.9/4.00

Nottingham, UK

Jun 2026

Honors: Outstanding Graduate (Top 5%, 2026); Dean's Scholarship (Top 10%, 2025); Outstanding Student (Top 5%, 2025); Zhejiang Provincial Government Scholarship (Top 1%, 2024); MCM Honorable Mention (2024).

Selected Coursework with 4.0/4.0: Computer Vision (ranked 1st), Computer Security (ranked 1st), Operating Systems & Concurrency, Algorithms & Data Structures, Artificial Intelligence Methods, Databases and Interfaces, Systems and Architecture.

SELECTED INTERNSHIPS

Ningbo AST Machinery Technology | github.com/asher0913/multi-material-slicer

Ningbo, China

Software Engineer Intern | C++17, Qt, OpenGL, CMake, Python, FastAPI, Vue.js, MySQL, LLMs

May 2026 – Aug 2026

- Built a cross-platform multi-material resin 3D-printing slicer in C++17, Qt and OpenGL: STEP/STL assemblies imported through OpenCascade with nested transforms preserved, per-part material assignment and a VBO-based 3D preview.
- Rasterized per-material PNG layer masks in a background worker with cancellation and timeouts, planned tank changes layer by layer and wrote merged exposures plus G-code; packaged the app for macOS and Windows.
- Added a headless --selftest that runs the full import-to-G-code path under Xvfb in Ubuntu GitHub Actions, alongside CMake builds and unit tests for the Python slicing backend.
- Designed a 51-table MySQL CRM (107 foreign keys, 70 constraints) and 10 Vue/FastAPI modules; built an LLM pipeline that produced 345 faculty profiles and 1,127 deduplicated grant records.

Zhiyang Innovation Technology

Shanghai, China

Machine Learning Engineer Intern | PyTorch, ONNX, TensorRT, NVIDIA Jetson, FastAPI, Docker

May 2025 – Aug 2025

- Cut Jetson Orin Nano inference latency from 220 ms to 95 ms (2.3×) and model size from 87 MB to 14 MB (6.2×) at under 0.5 mAP loss by exporting to ONNX and deploying INT8 TensorRT FP16 behind FastAPI and Docker.
- Raised mAP@0.5 from 0.74 to 0.79 and recall 6.2 points while cutting false positives 12% on a 7:1 imbalanced defect set; a training pipeline for 120K+ images cut label noise 18% and data preparation from 3 days to under 1, adding 5.6 mAP@0.5.

Beijing Corefire Technology

Beijing, China

Software Engineer Intern | Java, Spring Boot, Spring Cloud, MySQL, Redis, REST APIs

May 2024 – Jul 2024

- Implemented merchant onboarding and POS/QR query REST APIs for UMFintech in Spring Cloud and MySQL, plus reports that summarized onboarding volume, terminal activations and provider profit sharing by day, month or custom range; built token-based merchant login with sessions stored in Redis.

SIDE PROJECTS

InferenceLab | Python, asyncio, vLLM, LLM serving | github.com/asher0913/inference-lab

Sep 2024 – Oct 2024

- Built an LLM serving front end: an async dynamic batcher with batch-size and queue-wait bounds, single-flight coalescing of identical in-flight prompts and a TTL/LRU semantic cache over a swappable vLLM, SGLang or Ollama backend.
- In an 80-request, 16-concurrency load test on a deterministic backend, coalescing cut backend generations from 16 to 3 at an 80% cache-hit rate; GitHub Actions regenerates and checks the results.
- A labeled paraphrase study caught the semantic cache serving wrong answers (swapped unit conversions); a literal guard and per-family thresholds answered 67.6% of paraphrases with none wrong. A scheduling simulation showed continuous batching sustaining 10× the load of static batching within a 10 s p95 SLO.

CodeTaskForge | Python, FastAPI, Docker, pytest | github.com/asher0913/code-task-forge

Sep 2025 – Oct 2025

- Built a SWE-bench-style judge for coding-agent patches: path policy, throwaway workspaces, shell-free pytest runs with scrubbed environments and process-group timeouts, restored tests, fail-to-pass / pass-to-pass and held-out tests, shared by a CLI, a FastAPI service and Docker.
- On 98 labeled candidate patches across 10 tasks, exit codes accepted 25 wrong ones (test tampering, hard-coded answers), the SWE-bench protocol 15 and held-out tests none; each rejection got the correct one of eight failure kinds, and GitHub Actions reruns all 294 verdicts.

SecureRAG | Python, FastAPI, RAG, BM25 | github.com/asher0913/secure-rag

Oct 2023 – Nov 2023

- Implemented multi-tenant hybrid BM25 + embedding retrieval that applies tenant and ACL filters before ranking, computes collection statistics from the caller’s view only, and re-checks every candidate live against the directory of record, failing closed.
- Benchmarked five enforcement designs on a two-tenant corpus: post-filtering left 25% of contexts empty, and after permission changes a pre-filter on the stale index leaked on 54% of queries versus 0% with the live re-check; Recall@K, MRR and leakage checks run in CI.

SkillRouter | Python, LLM agents, embeddings | github.com/asher0913/skill-router

Aug 2026 – Sep 2026

- Three-tier skill registry with an IDF-weighted lexical prefilter and a ranker: on a 60-skill catalog, per-query context fell from 14,108 to 2,052 tokens (–85.5%) at top-1 accuracy 0.984 and top-3 1.000. On held-out paraphrases, lexical top-1 drops to 0.21, and an optional MiniLM stage lifts it to 0.56 while declining 85% of out-of-domain questions.

SELECTED PROJECTS

Oxford-IIIT Pet Classification | PyTorch, CNNs | github.com/asher0913/oxford-pet-classification

Nov 2024 – Dec 2024

- Fine-tuned an ImageNet ResNet-18 with differential learning rates, mixed precision, cosine scheduling, bias-corrected EMA and flip TTA to 89.8% accuracy on the official 3,669-image test split (37 breeds), selecting checkpoints on validation and scoring the test set once.
- Ran a 10-run progressive ablation of a from-scratch 2.78M-parameter SE-ResNet (35.2% → 49.7%): augmentation added 8.8 points and Mixup cost 1.3; analyzed errors with confusion matrices and a hand-written Grad-CAM.

Contactless Breathing Monitor | Swift, iOS, Vision | github.com/asher0913/contactless-breathing-monitor

Mar 2024 – Apr 2024

- Synchronized front-camera video with TrueDepth depth and used Vision shoulder landmarks to track the chest on-device; an autocorrelation estimator reports breaths per minute within 1 breath/min on 99.6% of synthetic signals at 60% noise. Grew out of a Software Engineering group project I led.

IAMABOT (MechMania 32) | Python | github.com/asher0913/IAMABOT

Sep 2026

- 24-hour AI contest bot commanding up to 32 units on a 32×32 map, with fire and payload control derived from the game engine’s Rust source and a replay-driven regression arena for testing strategy changes.

SELECTED RESEARCH

DualPath-CEM | Final-year thesis, PyTorch, privacy ML | github.com/asher0913/DualPathCEM

Nov 2025 – Jun 2026

- Designed dual-path split inference against model-inversion attacks: a Gaussian-protected 16×16×16 SlotCEM tensor carries the privacy burden, a 256-D MobileNetV3 token carries utility, and the server fuses two classifiers with calibrated logits; the attacker sees both released tensors.
 - On a 44,139-image FaceScrub split (526 identities, 5 seeds), reached 81.96% mean top-1, 1.63 points above the published Noise_ARL+CEM reference, while raising decoder and GAN reconstruction MSE 75.8% and 74.9%.
 - Released an audited snapshot of 70 formal attack runs and 15 utility repeats with a data manifest; a CI check verifies every reported metric against the frozen results.
- Manuscript in preparation: Y. Zhang and J. Ren, “DDP: A Disentangled Dual-Path Privacy-Protection Framework for Collaborative Face Inference,” Elsevier Pattern Recognition, 2026.

SELECTED EXTRACURRICULAR

Students’ Union, University of Nottingham

Nottingham, UK

Deputy Director

May 2023 – Apr 2024

- Organized a multilingual cultural season and film festival (400+ participants) and partnered with KPMG, NIO, Xiaomi, PwC, ByteDance and Lufthansa on career sessions, each engaging 200+ students.

School of Computer Science, University of Nottingham

Nottingham, UK

Peer Mentor

Sep 2023 – Jan 2024

- Mentored 20+ first-years in weekly Java, recursion, data structures, complexity and Git sessions; authored review sheets and earned helpful ratings from 90%+ of mentees.

TECHNICAL SKILLS

Languages: Python, C++17, Java, SQL, TypeScript, JavaScript, Go, Swift

ML / AI: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LLMs, RAG, AI agents, LoRA fine-tuning, embeddings, computer vision, NLP, OpenCV

ML Systems: vLLM, LLM serving, ONNX, TensorRT, INT8 quantization, MLflow, Docker, Kubernetes, CI/CD (GitHub Actions)

Backend & Tools: FastAPI, REST APIs, Spring Boot, MySQL, PostgreSQL, Redis, AWS, GCP, Linux, Git, Qt, OpenGL, CMake